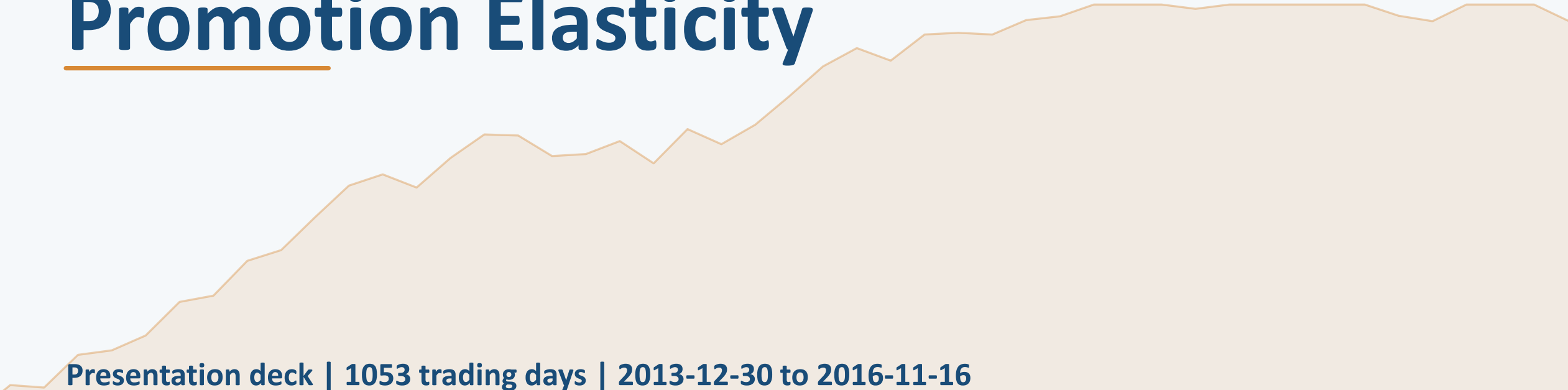


SALES CASE STUDY

Pricing, Profitability & Promotion Elasticity



Presentation deck | 1053 trading days | 2013-12-30 to 2016-11-16

Prepared for business analytics case study submission

Executive summary

Core findings from the sales, price and margin analysis

R186.9m

Total sales

Full analysis period

5.28m

Units sold

Total quantity sold

R35.4

Weighted avg price

Sales divided by total units

-3.81%

Margin pressure

Overall gross margin

Management message

Demand is highly price sensitive:

- ✓ lower selling prices generally stimulate stronger volume response.
- ✓ Promotions increase basket volume but do not automatically improve profitability.
- ✓ Margin performance is structurally thin, so promotion depth and cost data require close control.
- ✓ The pricing strategy should optimise profit contribution, not only revenue or units sold.

Decision focus

Use promotions selectively. A campaign should be approved when expected volume uplift is large enough to offset discount cost and protect gross profit.

Analytical approach

From raw daily records to decision-ready metrics

01

Validate

1053 daily records,
no missing gaps or
nulls, sales, cost
and quantity
reconciled.

02

Derive

Calculate unit
price, unit cost,
gross profit, GP%
and GP% per unit
per day.

03

Detect

Flag promotions
where selling price
falls below a rolling
baseline.

04

Explain

Measure elasticity
against matched pre
and post promotion
baselines.

Additional analysis included in this deck

- ✓ Regular vs promotional trading comparison
- ✓ Performance risk items: negative GP days and promotional days
- ✓ Decision scorecard for pricing and promotion governance

Headline numbers

The full period at a glance

R186.9m

Total sales

5.28m

Units sold

R35.4

Avg unit price
weighted

-3.81%

Overall gross
margin

R37.07

Avg daily price
unweighted

R30.7 - R43.81

Price range

607 of 1053

Negative GP days

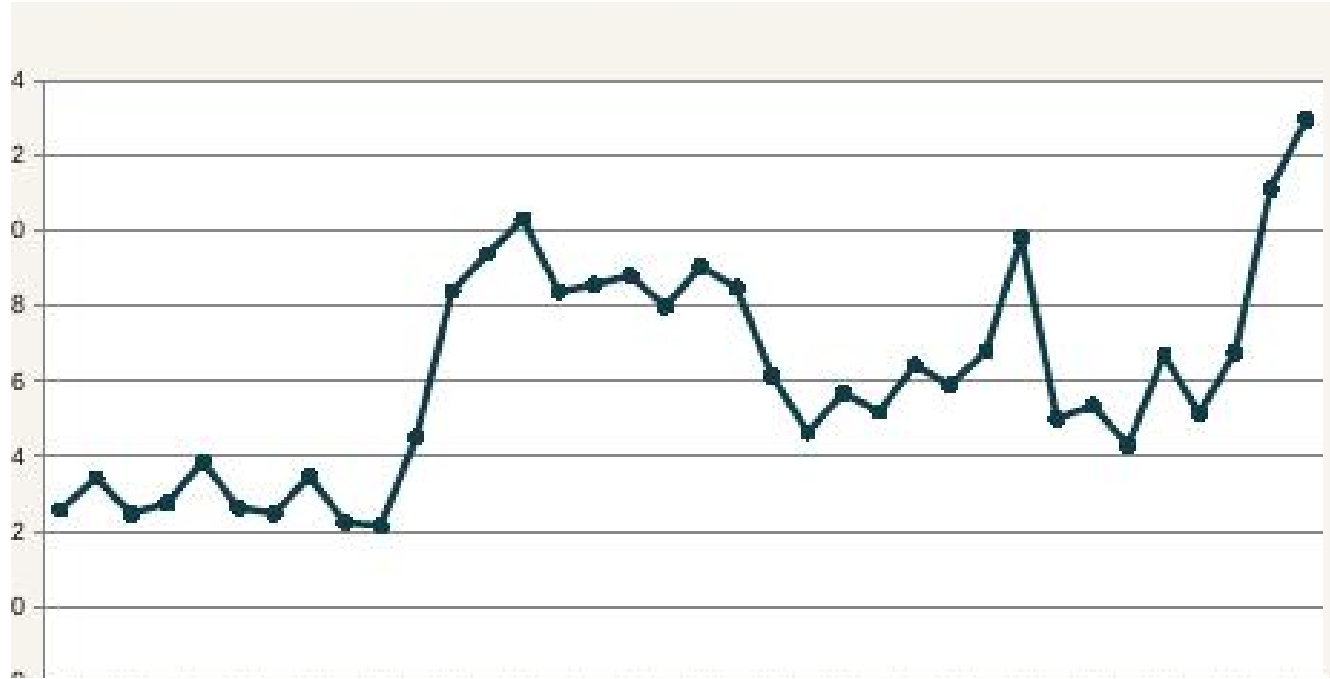
104

Promotion days

Interpretation: the product generates meaningful revenue and volume, but headline profitability is negative. This means pricing and promotion decisions should be assessed through margin contribution, not only sales growth.

Unit selling price

Daily price equals Sales ÷ Quantity Sold. Monthly view shown below.



Answer

Weighted average unit price: R35.4
(total sales ÷ total units)

Simple mean of daily prices: R37.07

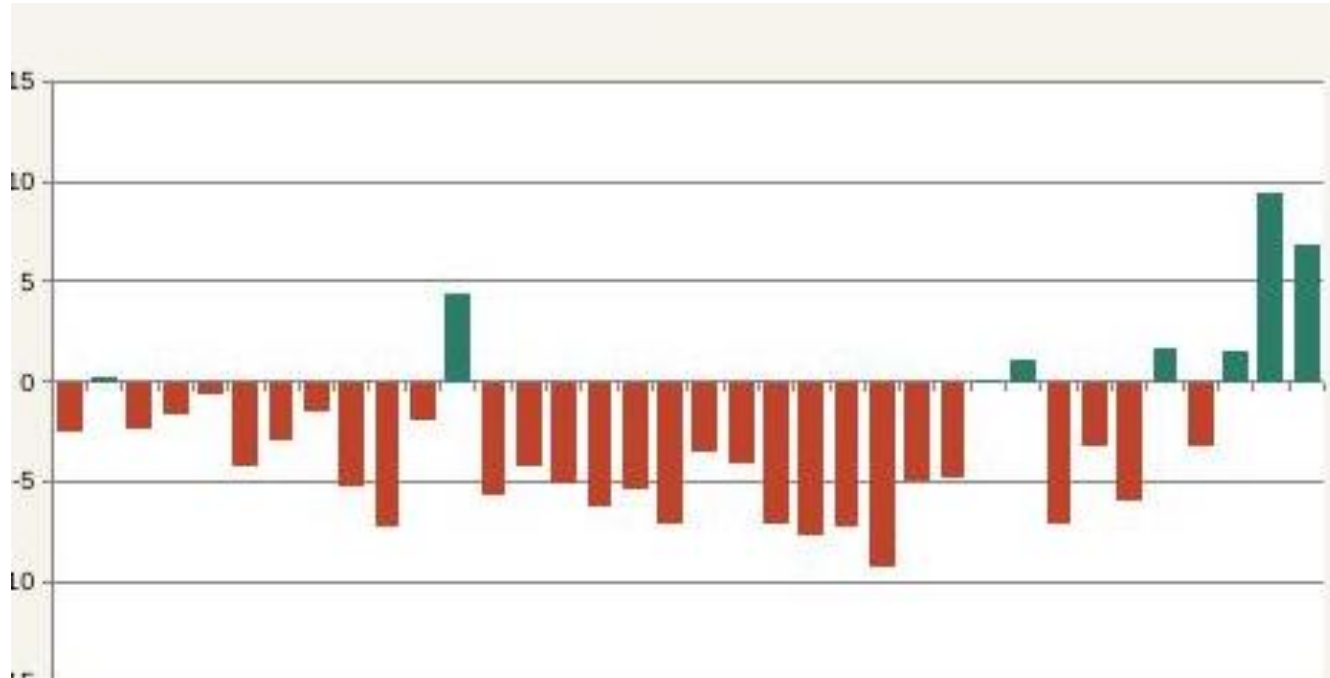
Median: R36.91

Interpretation

Price drifted upward from about R32 in 2014 to above R42 late in 2016, with sharp promotional dips. This supports active price management and frequent discounting.

Daily gross profit %

$$\text{GP\%} = (\text{Sales} - \text{Cost of Sales}) \div \text{Sales}$$



Overall margin: -3.81%

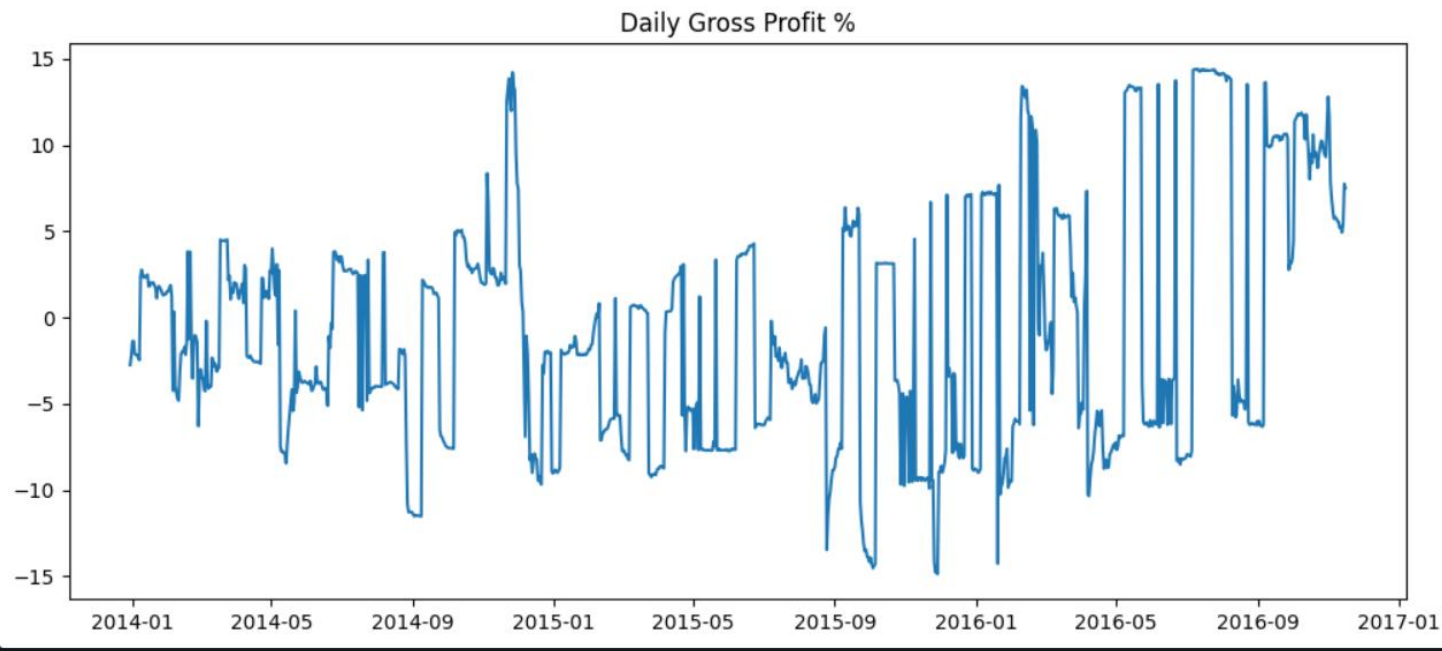
Average daily margin: -0.87%

GP% per unit is mathematically identical to daily GP% because both reduce to price minus unit cost divided by price.

607 of 1053 days, 58%, sold below cost of sales. This is the most important profitability risk in the case study.

Additional view: Daily gross profit volatility

Daily GP% highlights how frequently margins cross into negative territory



Interpretation:

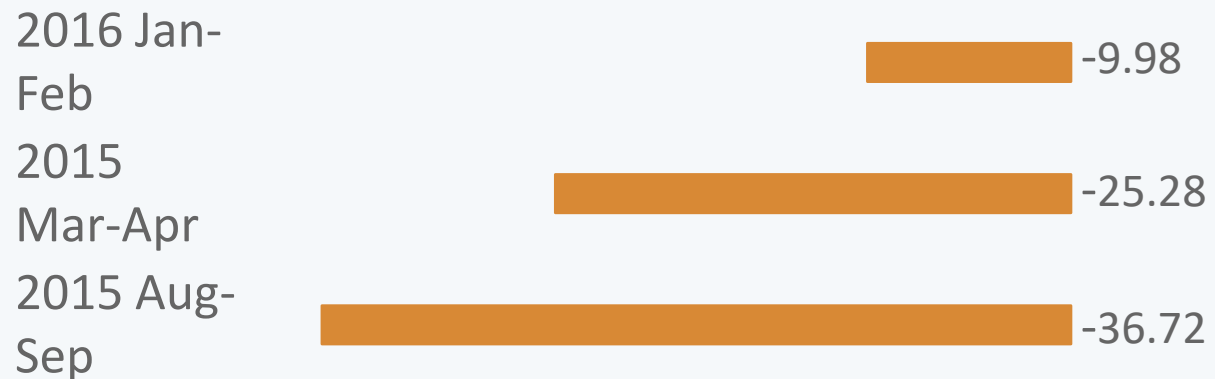
The plot fluctuates between roughly -15% and +15%. Persistent negative periods indicate that some pricing decisions or cost movements eroded profitability. The improvement toward 2016 suggests pricing discipline or better cost recovery.

Three promotional periods

Price elasticity of demand using matched pre and post promotion baselines

Period	Days	Base to promo price	Δ Price	Δ Units	PED
2016-01-25 to 2016-02-08	15	R40.25 to R34.04	-15.42%	+153.98%	-9.98
2015-03-24 to 2015-04-06	14	R40.58 to R37.75	-6.98%	+176.51%	-25.28
2015-08-25 to 2015-09-07	14	R38.75 to R34.23	-11.65%	+427.79%	-36.72

Elasticity comparison



Conclusion

All three promotional periods are highly elastic. A 1% price reduction generated a much larger percentage uplift in units sold. The strongest volume response occurred in the Aug-Sep 2015 promotion.

Caution: elasticity confirms demand response, not profit improvement. Margin must be evaluated separately.

Does the product perform better on promotion?

Volume says yes. Profit says no.

Regular price

Avg daily sales	R167,664
Avg daily units	4,720
Avg unit price	R37.27

Avg daily GP R-5,148 (-0.18%)

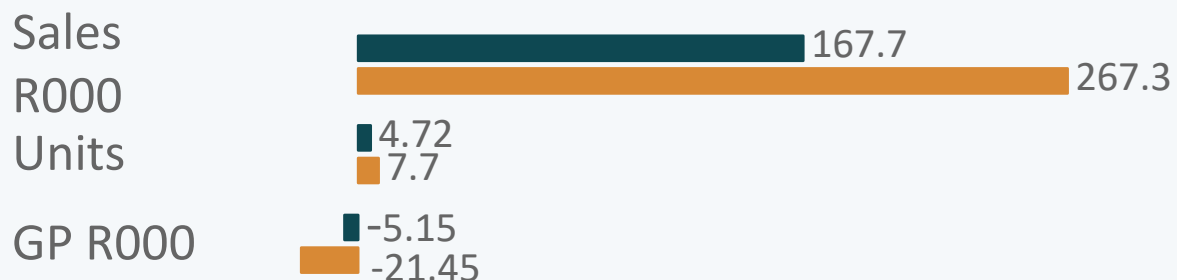
Promotional

Avg daily sales	R267,277
Avg daily units	7,702
Avg unit price	R35.24

Avg daily GP R-21,453 (-7.24%)

Comparison chart

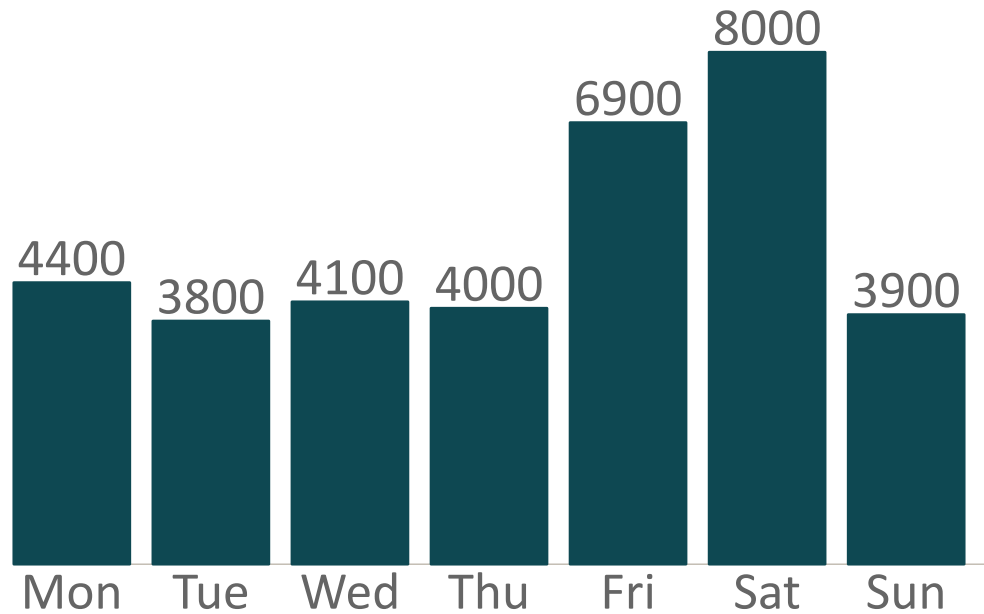
Regular Promotion



Verdict: promotions win share of basket but reduce margin. Daily units rise about 63%, while average daily gross profit drops from R-5.1k to R-21.5k.

Other insights

Trading rhythm and structural margin risk



Weekend engine

Friday and Saturday carry about 44% of weekly units at almost identical prices, so volume is not discounting driven.

Margin is structurally thin

Average margin sits near zero and turns negative on many days. The risk is below-cost selling.

Promotion clusters

Aug-Oct 2015 and Jan-Feb 2016 hold the deepest discount blocks and largest volume spikes.

Promotion governance scorecard

A practical way to turn the case study into management action

Negative GP days

607 / 1053

58% | Profit risk

Promotion days

104 / 1053

10% | Campaign exposure

Max unit response

+428%

High | Elastic demand

Suggested promotion rule

- 1 Set a price floor** No campaign should be approved below the margin-safe price floor.
- 2 Forecast unit uplift** Use PED by campaign to estimate volume response.
- 3 Estimate gross profit** Approve only when contribution improves versus base case.
- 4 Track actuals** Compare expected and actual unit uplift, sales and GP after each campaign.

Recommendations

Translate analytical findings into pricing and promotion decisions

1 Re-base the promotional price floor

Discounts below about R34 push unit economics negative. Cap discount depth and shorten promotion duration.

2 Audit the cost of sales feed

Cost exceeding revenue on 58% of days points to costing, allocation or supplier cost pressure.

3 Trade the weekend, not only the price

Weekend demand is naturally strong. Shift promotional spend toward weaker days and protect full-price weekend volume.

4 Track elasticity per campaign

Adopt the PED baseline method used here as a standard post-campaign scorecard.

5 Add margin guardrails to every campaign

Evaluate promotions on gross profit contribution, not total revenue alone.

Closing message: the product performs strongly in volume terms, but pricing discipline is required to convert volume response into sustainable profit.